QP CODE: 25900649

Scheme for Valuation

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third Semester

Discipline Specific Core Course (DSC) – MG3DSCSTA201

ANALYTICAL TOOLS FOR MULTIVARIATE ANALYSIS

2024 ADMISSION ONWARDS

Duration: 1 Hour 30 Minutes **Max Marks: 50**

Part A

Answer any 7 questions (Each question carries 2 marks)

1. Statement of Basis theorem (2 Mark)
2. Definition of orthogonal vectors (2 Mark)
3. Definition idempotent matrix with example (2 Mark)
4. Definition of row echelon form and reduced row echelon form (2 Mark)
5. Definition of orthogonal matrix with example (2 Mark)
6. Definition of trace of a matrix with example (2 Mark)
7. State norm (1 Mark)
   Triangular inequality of norm (1 Mark)
8. Any 2 applications of linear algebra in statistics (2 Mark)
9. Condition for diagonalizable matrix. (2 Mark)
10. State the relation of eigen vectors (2 Mark)

Part B

Answer any 4 questions (Each question carries 6 marks)

11. a) Definition of Scalar and Vector with example (2 Mark)
    b) Definition of Norm and its Significance (1 Mark)
    c) $\cos\theta = 0.272$ and $\theta = 1.295$ (3 Mark)
12. Proof (6 Mark)
13. Determinant of A and B (3 mark)
    Determinant of AB (3 Mark)
14. Computing eigen value ($\lambda 1 = -3, \lambda 2 = -2\sqrt{3} - 3\,, \lambda 3 = -3 + 2\sqrt{3}$) (4 Mark)
    Result (2 Mark)
15. a) Definition of Cramer's rule (2 Mark)
    b) x= -3/5, y=1/5 , z=12/5 (4 Mark)
16. Statement of Cayley Hamilton theorem (2 Mark)
    Characteristic Equation (1 Mark)
    verification of theorem (3 Mark)

Part C

Answer any 1 question (Each question carries 12 marks)

17.	Definition of linear independence of vectors	(2 mark)
	Linear combination of vectors and equating	(1 mark)
	Result	(9 mark)
18.	Computation of R($\pi/4$)	(3 mark)
	Computation of R($\pi/2$)	(3 mark)
	Computing R($\pi/2$)S R($\pi/2$)	(6 mark)